

RND-6: SAFE IN-TIME COVARIATES ??? CUPAC-STYLE ????????

> ??????: ? ??????. ??????????? ?????????????? ?????????? ? ??????? ???????
> A/B ? CUPAC A ? A+B ? A+B+C; ?????????? ?????????? open-source ?????????? ? ?????????? ???.
> ?????????? ??????? ?? ??????? ? notebook.ipynb (??????????, seed=11).

????

?????????, ??? ? ? ?????????????????? ?????????? ?????????? in-time ?????????? ??????
scikit-CUPAC ?????????????????? ?????????? ?????????? (? ?????????????? ?????????? ??????????) ???
????????????? ?????????????????? ?????????????? ??????? ATE.

?????? ?????????? (SAFETY LAYER)

? **A** pre-treatment ? ??? ? CUPAC;
? **B** expert-safe in-time ? ?????????? second-stage ??????????;
? **C** balance-gated in-time ? ?????????? ?????????? ????? balance/missingness gate;
? **D** dag-required ? ??? estimator (????? causal-?????????);
? **E** mediator-risk, **F** leakage ? ?????????? ??? ?????????? ????? ATE;
? **unsafe_demo** ? ?????? ?????????????? ??????, ?? ??????????

??????

ab_hypex (A/B baseline) • hypex_cupac (reference/parity, ??? ???????) •
sklearn_cupac_A ? sklearn_cupac_A_plus_B_linear ? sklearn_cupac_A_plus_B_plus_C_linear
(????????? ?????????? ??????????) • unsafe_demo_optional (??????????????).

????????????? ?? ?????????? (N=8000, SEED=11, ?????????? ATE = 2.8)

method	ate	p_value	variance_reduction_vs_ab_%	incremental_vs_predecessor_%	safety_status
---	---	---	---	---	
ab_hypex	2.758	0.0	0.0	?	ok
hypex_cupac	2.785	0.0	13.56	?	reference_only
sklearn_cupac_A	2.801	0.0	25.63	25.63	ok
sklearn_cupac_A_plus_B_linear	2.775	0.0	42.38	22.52	ok
sklearn_cupac_A_plus_B_plus_C_linear	2.766	0.0	52.80	18.07	ok
unsafe_demo_optional	1.000	0.0	83.17	?	unsafe_demo

(????????? ?????????? ?????????? ? ??????? ?????????????? ?????????? ?? ?????????? ??????????)

????????? ???????

1. ?????? ?????????????? ??????? ??? ? ?????????????????? ?????????: A (~26%) ? +B (~42%) ? +B+C (~53%)
????????? ?????????? ?????????????? A/B, ??? ??? ATE ?????????? ?????????????? (? 2.8).
2. CUPAC parity: hypex_cupac (~14%) ?????????? ?? ?????? ? sklearn_cupac_A;
????????????? ?????????????? ??????? ??? R&D ? ?????????????????? ??? ??????????
3. Balance/missingness gate ?????????????? A/B/C (|SMD|?0) ? ?????????? D/E/F ? unsafe_demo
(|SMD| ? 0.45?1.14, p?0) ? ?????????, ?????? ?? ?????? ?????? ? estimator.
4. unsafe_demo ???? ?????????????????? «????????? ??????????» (~83%), ?? ????????? ATE ? 0 ?

outcome-derived ; ? outcome-derived.

balance-test gate (E/F), ** second-stage, ML-

HYPEX

HypEx CUPAC.
HypEx: (1) B/C open-source;
(2) balance/missingness gate ; (3) E/F unsafe_demo. CUPAC
VarWar (~34% ~33%; ,).

bash
pip install -e .[ml]
?. NOTEBOOK.IPYNB; ATE-CI ? RND_REPORTS.BENCHMARK
python tools/generate_pdf.py # report.pdf